The Data Card answers

Input vs. Output

Users (smallholder farmers or extension workers) input a short, natural-language question in text form — for example, "Why are my maize leaves turning yellow?" The model's output is a ranked list of the top-5 most relevant documents from the agricultural extension knowledge base, ordered best-first. The system does not generate the final farming answer; it retrieves and ranks the source factsheets that a downstream RAG answer would rely on.

Classification problem

This is fundamentally a *ranking/retrieval* task rather than a classification task, but it uses graded relevance labels as its supervision signal. Each (question, document) pair is assigned one of four classes:

- 3 — Perfect: directly answers the question
- 2 — Relevant: a closely related aspect of the same topic
- 1 — Marginal: the same issue on a different crop
- 0 — Not relevant (including hard negatives — documents that look similar but are wrong, e.g. potassium-deficiency advice for a nitrogen question)

These grades were chosen because a farmer's question usually has several partially-relevant documents, so ranking *order* matters — the system must place a "perfect" document above a merely "marginal" one.

Labeling

Labels come from expert relevance judgements provided with the competition dataset (agricultural domain knowledge). In a real deployment, we would source labels from agronomists and experienced extension officers rather than crowd workers, because distinguishing a correct diagnosis from a plausible-but-wrong one requires domain expertise. Consistency would be checked by having multiple experts label overlapping samples and measuring inter-annotator agreement, resolving disagreements through review.

Data creation

The dataset is a self-contained knowledge base of 695 short agricultural extension factsheets covering crop diseases, pests, nutrient deficiencies, soil management, climate adaptation, and fertiliser advice, paired with farmer-style questions and graded relevance labels. The corpus mixes original factsheets with rewrites of real CC-BY agricultural research, so vocabulary and phrasing vary. For a real low-resource deployment, we would extend it by collecting genuine farmer questions from extension services and translating/localising factsheets into local languages.

Web collection ethics

The competition corpus is self-contained and built partly from CC-BY licensed research, which permits reuse with attribution — so copyright is respected at source.

If we extended the dataset from the web, we would restrict ourselves to openly-licensed agricultural sources (e.g. government extension services, CGIAR, FAO), preserve attribution, and avoid scraping personal or identifying farmer data. Any real farmer questions collected would require informed consent and be anonymised to protect privacy.

Bias and representation

Groups at risk of exclusion: farmers growing minor or non-staple crops (the corpus may over-represent maize and major staples), farmers who phrase questions in local languages or dialects rather than standard English, and those asking short, conversational, or grammatically imperfect questions. We will measure retrieval quality *separately across crops and query types* to detect uneven performance, and deliberately include short/informal queries and under-represented crops so the system does not only serve well-phrased, staple-crop questions.

Values and community alignment

The dataset design reflects the Challenge 1 values: accuracy and relevance (graded labels and hard negatives force the system to distinguish true intent from surface keyword overlap), inclusivity (testing on informal and short queries), fairness (per-crop evaluation so no group is systematically underserved), and transparency (retrieval traces every answer back to a sourced factsheet). The community benefits through more reliable agricultural guidance; accountability is maintained by keeping guidance traceable to original documents and by involving domain experts in labelling and review.